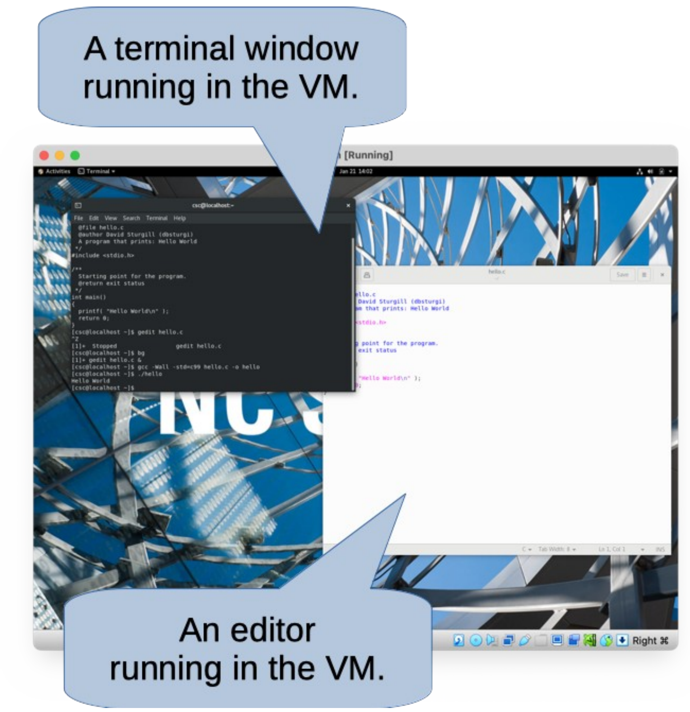
Using a Virtual Machine

We're making a pre-built Centos 8 Virtual Machine (VM) available. This is one option way you can work on the exercises and projects. This should work if you're on an Intel-based or AMD-based Microsoft Windows or MacOS system. If you're on a newer system that uses an ARM processor (e.g., the M1, M2, etc MacOS systems), then I don't think these instructions will work for you. I have successfully used a Linux VM on my ARM MacOS system, but I don't have a ready-made VM image to give you.

Using a Linux VM is like running a Linux machine inside a window on your desktop. Inside that window, you can install and run all applications you might normally expect to use on a Linux machine. The following picture shows what using this machine will look like.



The Centos 8 distribution run by this VM should be fairly close to the configuration used by university EOS Linux machines. We've also tried to make sure it has the various developer tools we'll need during the semester.

Installing the Virtual Machine

If you want to try out the image, you'll need to download and install a copy of Oracle VirtualBox for your system. It's free and available at: https://www.virtualbox.org/ Visit this site and install a version for whatever operating system you're using (probably Windows or MacOS). I used version 7.0.6 of the VirtualBox software when I created the Centos 8 image, but it should be OK if the version you get isn't exactly the same.

After installing the VirtualBox software, you can download the Virtual Machine image using the link on our Moodle page. The image is a file named bean.ova, and it looks like it's about 4.7GB to download, but it will take about 9.5 GB after you import it.

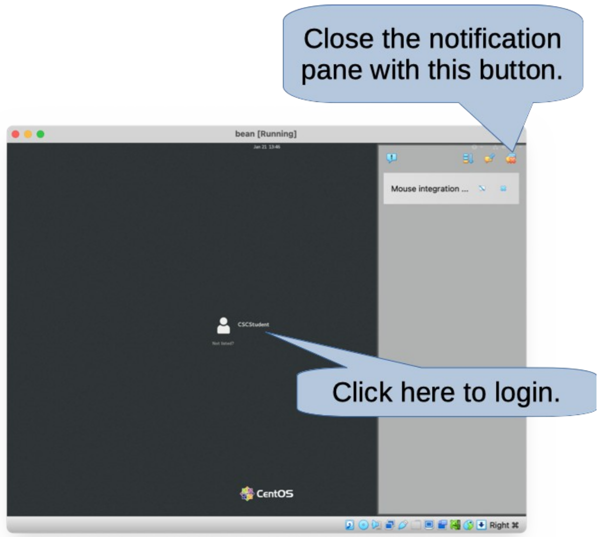
After downloading the bean.ova file, visit the VirtualBox application and you should see an "Import Appliance" item under the File menu. Choose this and select the bean.ova file you downloaded. You should be able to take the default options to expand this file into a virtual machine that's ready to run.

In VirtualBox, select the virtual machine named bean over on the left side, then click on the green arrow near the top of the window to boot up the VM. This will take about a minute. In the future, you can make this faster by by suspending your VM when you don't need to use it (rather than shutting it down and re-starting every time). This will let you resume it quickly, and it will start back up right where you left it the last time you were using it (same windows running, etc).

Using the Virtual Machine

After the virtual machine boots, you should be able to login as the user named "CSCStudent" (username csc). You will probably get a notification pane over on the right side of the screen. There's a button up near the top to close this.

This csc user should let you login without a password. Feel free to add a password if you want to or create another user. The csc account is on the sudoers list, so you can run commands as root inside the VM when you need administrator privileges, say if you want to install more software or create a different user for yourself.



Inside the VM, you should be able to run a terminal window (or, several of them). Choose the **Activities** menu near the top and then click on the little terminal icon over on the left. From the terminal, you should be able to run text editors like nano, vim, gedit or emacs. The compiler is already installed, along with (I think) every other development tool we'll use this semester.

When you're done using the VM, just close the window and it will give you the option of suspending and saving the machine's state. Suspending the machine will take a little more disk space, but it will let it start up a little faster the next time you want to use it, and it will restore the machine just the way you left it. If you want to shut it down completely, you can do that from inside the VM, just click on the little triangle in the upper right corner, then click on the icon that looks like a power button.

Sharing Files

You can easily share parts of your host file system with your virtual machine. Then, you can use your VM to access the same files you'd normally work with on your laptop or desktop system. That's what I do when I'm working with the VM. I keep very little on the VM itself. Instead, I use it as a way to access files on my laptop via another operating system.

If you want to share files, you need to tell VirtualBox what to share. From virtual box, choose the menu item, **Device → Shared Folders → Shared Folder Settings...** A dialog box should pop up. Click the little folder plus icon over on the right. You can choose a directory on your host to share with the VM. Then, everything under that directory (and all the sub-directories) will be visible inside the VM. Once you share a folder, you can reboot the VM to get it to show up (I didn't really have to reboot my VM to see the shared folder, but you could try this if it doesn't show up automatically). If you mark the shared folder as permanent when you set it up as a shared folder, it should be there every time you start up the VM.

By default, your shared folders will show up under a path like /media/sf_folder in the virtual machine. You can create a symbolic link to a path in your home directory, making it easier to get to files on the host from inside the VM. For example, if I keep files under a directory named "class", I can share that with my VM and have it show up as /media/sf_class. Then, in my home directory on the VM, I can run the following command to create a shortcut to this path, so it's easy to get to files under my class folder either from my host or from inside the VM.

```
ln -s /media/sf_class ~/class
```

Networking

From inside your VM, you should be able to use the host's network. In the past (for older versions of the VM), some students have reported that the networking isn't activated automatically when the VM starts up. If this is the case, you can try clicking on the little triangle icon in the upper right corner of the VM window. The menu that pops up has an entry for your network connection. It should let you activate your network connection if it isn’t already activated. The VM will think it’s using a wired connection, even if your computer is really using wireless connection to talk to the outside world.

Screen Resolution

If you want more desktop space in the VM, you should be able to enlarge or full-screen the window, and the linux VM inside should grow to fill the space. You may sometimes have to select the menu item View –> Adjust Window Size, to get it to repaint after a resize. On some hosts (e.g., MacOS with retina display), the screen resolution may be too high to read the text in the VM. There’s an option for scaling the VM screen to help correct for this. You can select the menu Machine –> Settings …. Then, choose the Display tab, and use the Scale Factor to enlarge the rendering of the VM window.

Installing Packages

You should be able to install additional software in the VM. You can use yum to install packages from the command line, or you can choose the “Software” program from the “Activities” menu up near the top.

Other Topics

I'll add more to this document as people run into problems or questions.

I'll try to use the VM in class one day to show our examples, so you can see if it's worth trying out.